

Inpatient Hyperkalemia — 5-Step Framework

Confirm → ECG → Stabilize → Shift → Eliminate · Any K >5.5 gets a STAT ECG; treat K >6.5 as emergent regardless of ECG · complications: severe weakness, fatal arrhythmias

1 · Confirm — without delaying treatment

- Expected? (AKI/CKD/ESRD, RASi, K-sparing diuretics, TMP-SMX, rhabdo, TLS, acidosis)
- Hemolysis, fist clenching, poor collection → spurious K
- Symptoms: myalgias, weakness, palpitations

2 · STAT ECG — progressive changes

Early	Peaked T waves, QT shortening
Worse	PR prolongs → P flattens/lost; QRS widens
Pre-arrest	Sine wave, VT/VF, asystole; pseudo-RBBB / pseudo-STE V1–V2

- Repeat ECG q30–60 min until normal; telemetry
- ECG changes predict outcome better than the K level, but correlate poorly with it; **rapid rise >> chronic elevation** for arrhythmia risk

3 · Stabilize myocardium — ECG changes or K >6.5

	Ca gluconate	Ca chloride
Dose	1–2 g IV, peripheral OK (preferred)	0.5–1 g IV, central line (caustic)
Kinetics	Onset 3–5 min · lasts 30–60 min	
Effect	No change in serum K — should normalize ECG; re-dose q5–10 min if changes persist	

4 · Shift K into cells — temporizing only

Agent	Dose	Onset/lasts	ΔK (mEq/L)
Regular insulin + D50	10U IV + amp D50; 5U if renal impairment ; no D50 if BG>250	15–60 min / 4–6 h	↓0.5–1.0
Albuterol neb	10–20 mg in 4 mL NS (>> 2.5 mg asthma dose)	15–30 min / 2–4 h	↓0.5–1.5
NaHCO ₃	50–100 mEq IV — only metabolic acidosis, not ESRD	15–30 min / 1–2 h	least effective

- Monitor glucose ~6 h post insulin (hypoglycemia = most common complication)
- Albuterol: caution active CAD; best *with* insulin
- Pick cocktail by comorbidity; **shifting never removes K**

5 · Eliminate K — kidney ~90% / gut ~10%

Therapy	Use
Furosemide 40–160 mg IV	1st-line if hypervolemic; + isotonic IVF if euvolemic; useless in anuria ; alone can volume-deplete → worse AKI
HD / CRRT	Severe renal dysfunction or refractory — most likely to normalize K within 4 h
SZC (Lokelma) 10 g TID ×48 h	Onset ~1 h, preferred binder; Na load/edema, cost. UKKA 2023: option alongside shifting in life-threatening hyperK (K≥6.5)
Patiromer	Colon only, onset ~7 h — chronic niche
Kayexalate (SPS) 15–60 g q6h PO / 30–50 g PR	Avoid: post-op, ileus/SBO, IBD, infectious colitis (bowel necrosis); separate other meds 3 h; skip if HD planned. UKKA 2023: not for emergency tx

Do / AVOID

- **DO** recheck K in 1–4 h; repeat shift + eliminate until K normal or HD
- **DO** (KDIGO 2024) lower K with diet review/diuretic/binder **rather than stopping ACEi/ARB** — RASi d/c linked to ↑mortality/CV events in CKD
- **AVOID** NaHCO₃ in ESRD (HD pts were the non-responders) or without acidosis
- **AVOID** Kayexalate with any bowel risk or imminent HD
- **AVOID** judging calcium by the K level — its endpoint is the ECG

Responders to bicarbonate in studies: K>6, HCO₃<17, pH<7.35, not on HD (Abuelo, KI Rep 2018).